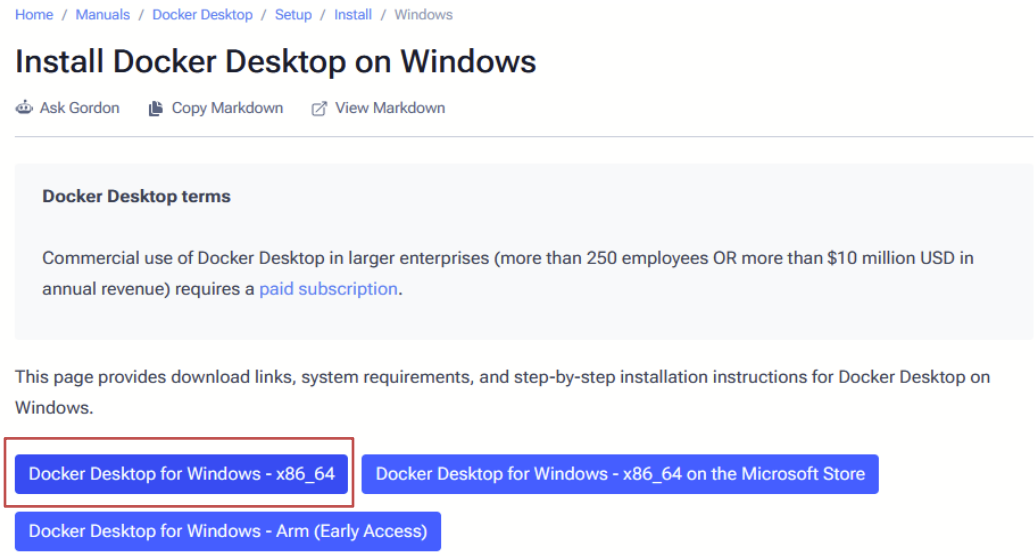
# INSTALLATION AND RUNNING MicroICS

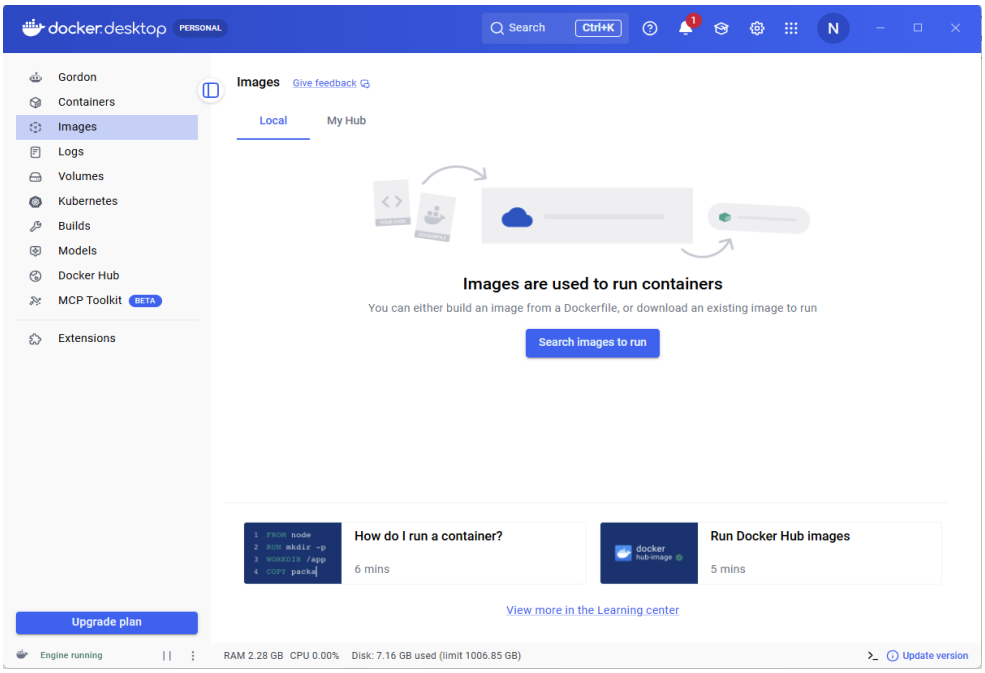
MicroICS is a tool to analyse 2D or 3D biofilm images. It runs as a self-contained Docker image (https://hub.docker.com/r/skblaz/microics), accessed through a browser-based interface, and the source code is available on GitHub (https://github.com/SkBlaz/biofilm-classification).

## Installation on Windows

- **Install Docker Desktop**
  - On the Docker download page (https://docs.docker.com/desktop/setup/install/windows-install/), click the button for your version of Windows (amd64 or arm64). The download will start.




  - When it finishes, double-click the downloaded file to begin installation.
  - Follow the installer, using the default (recommended) options.
  - Once installed, start Docker Desktop and wait until the whale icon indicates that it is running. Active Docker Desktop looks like this:



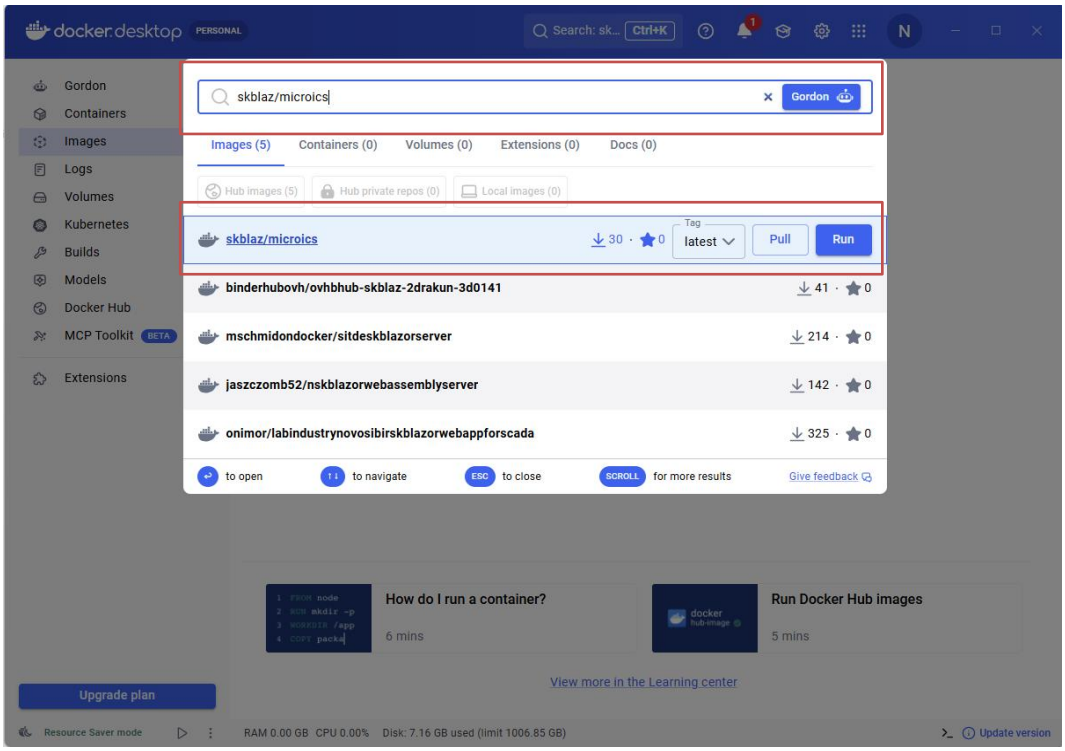
- **Setting up MicroICS (via Docker)**

You can do this in two ways: using Docker Desktop (the graphical interface), or using the command line (PowerShell on Windows, Terminal on macOS/Linux).

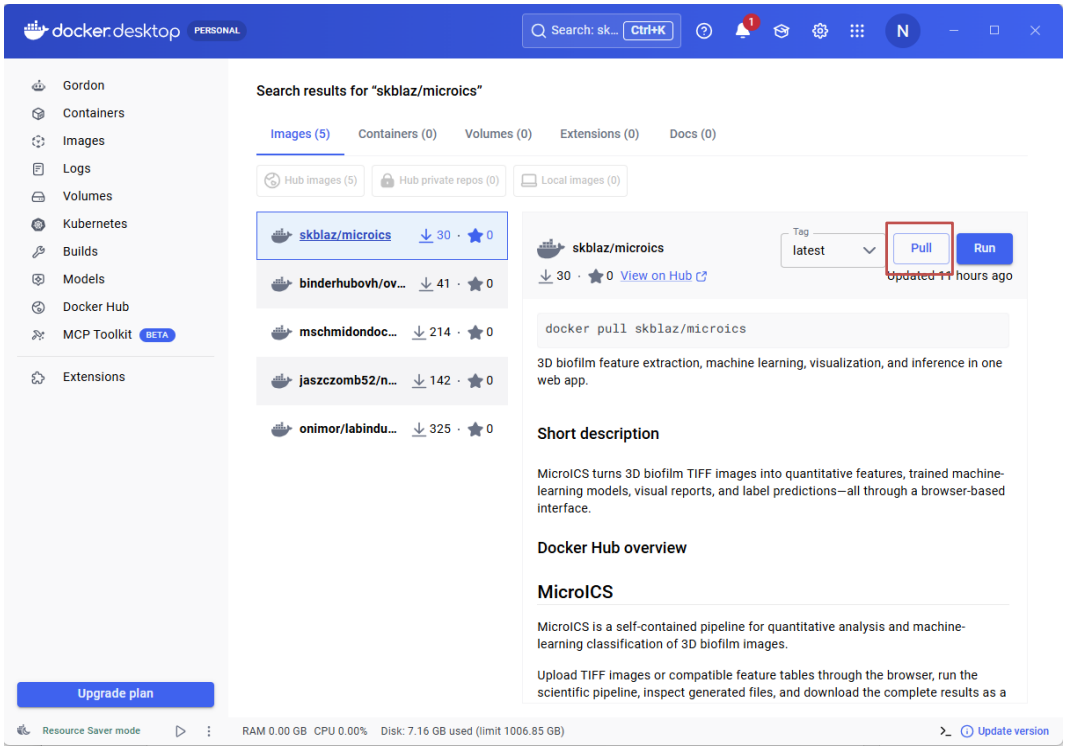
Before Docker is installed and run, WSL, the virtual machine, and Hyper-V may need to be activated on Windows. You can do this by searching for “Windows Features” and selecting “Turn Windows features on or off”. Tick Hyper-V, Virtual Machine Platform, and “Windows Subsystem for Linux”. Click OK and restart the computer, then continue installing Docker. If other issues arise, please resolve them with your IT department.

**Option 1 — using Docker Desktop (graphical)**

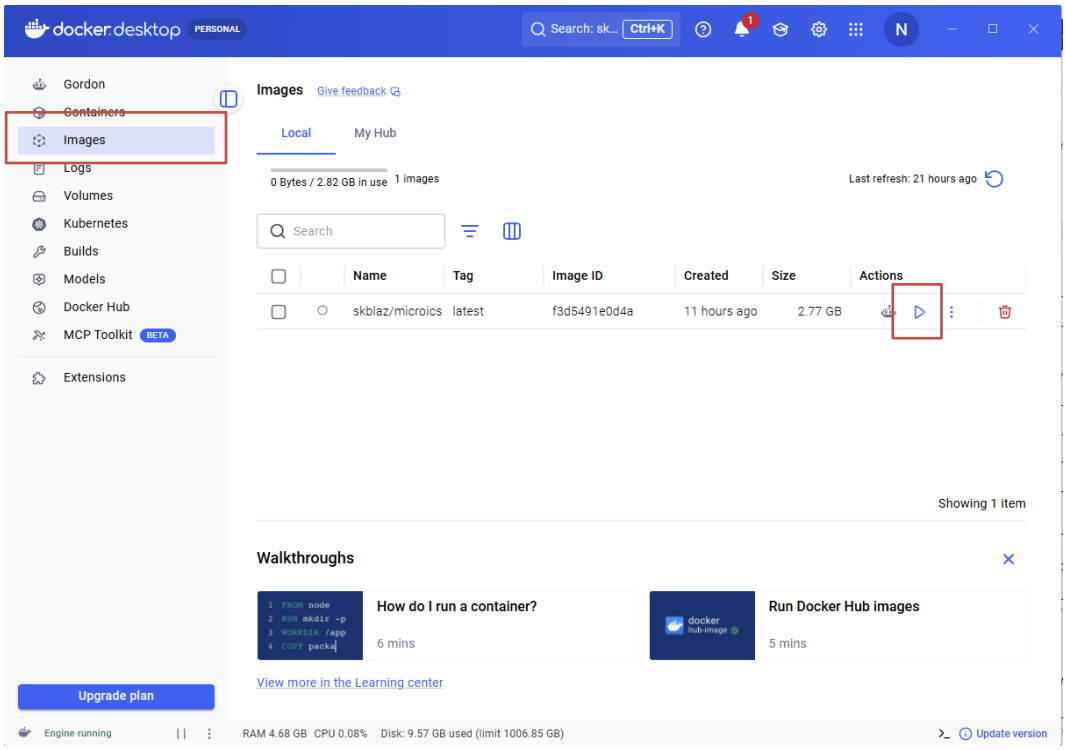
- Start Docker.
- In the search bar at the top of Docker Desktop, type **skblaz/microics**.



- Select it by clicking, then click **Pull** to download it. This may take a few minutes.

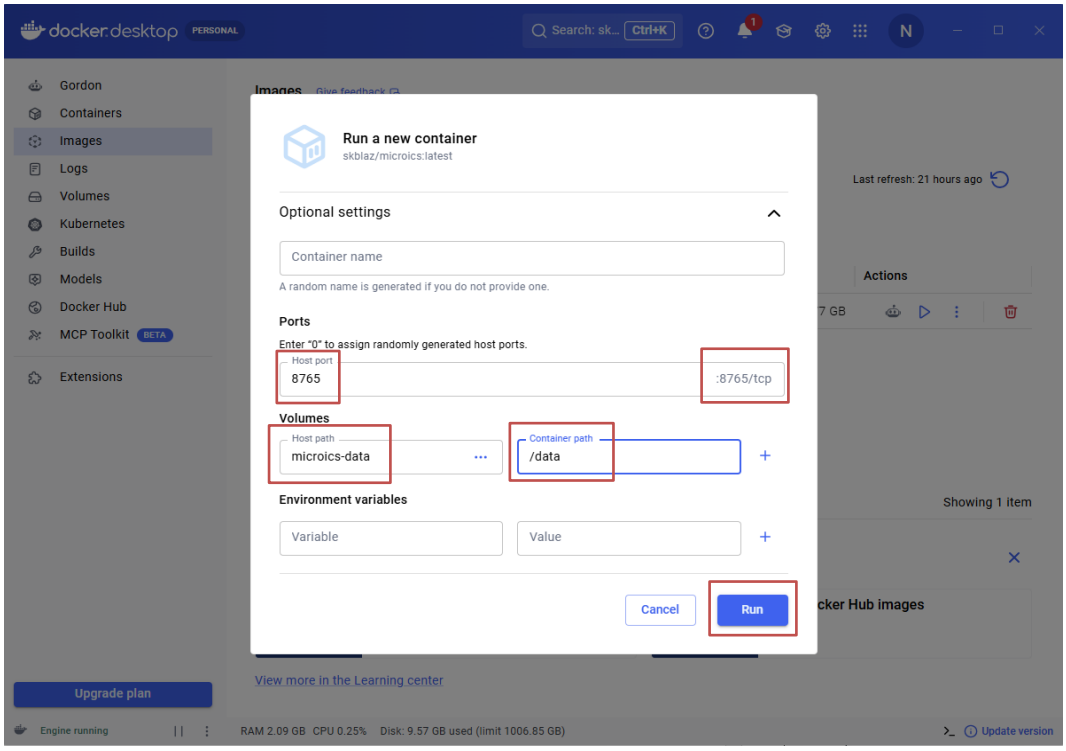



- When it has downloaded, open the **Images** tab on the left. Find skblaz/microics and click **Run** (the ▶ arrow on the right).

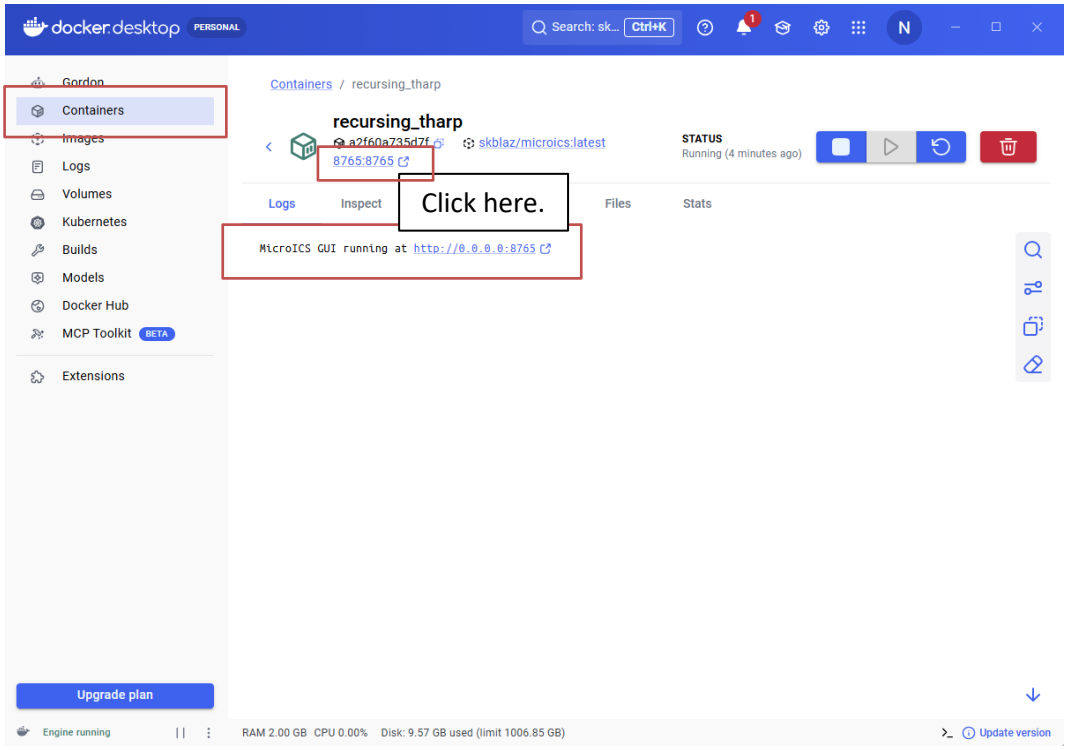


- In the dialogue that appears, click **Optional settings** to expand it, and enter:
  - **Ports** — Host port: 8765 (Container port: 8765, in grey, usually automatically filled)
  - **Host path**: microics-data, **Container path**: /data

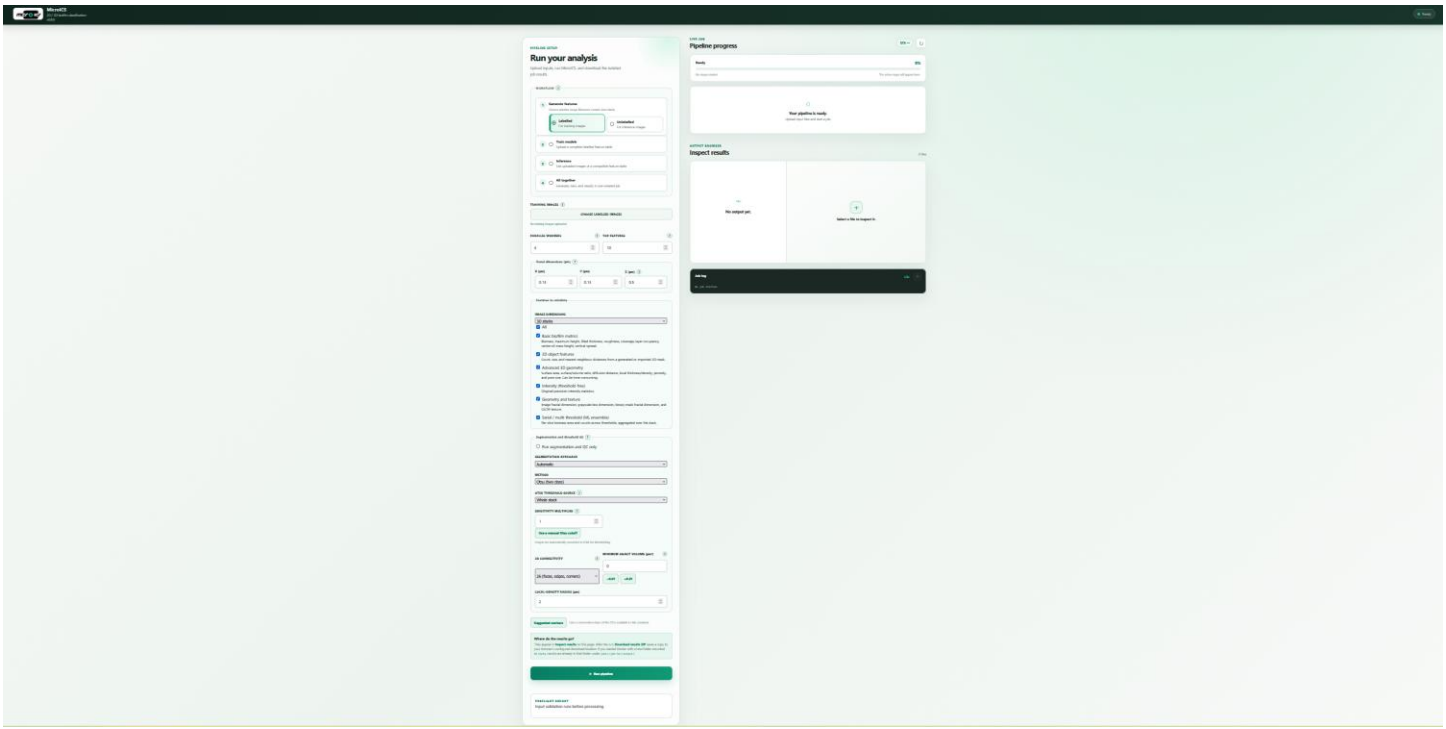
After you enter these, click **Run**.




- The container now appears under the **Containers** tab. Docker Desktop shows a clickable **8765:8765** link next to the running container or above it.



- The MicroICS interface should look like this:




**Option 2 — using the command line**

- Make sure Docker Desktop is running, then open PowerShell (Windows) or Terminal (Mac/Linux).
- Download the image:

```
docker pull skblaz/microics:latest
```

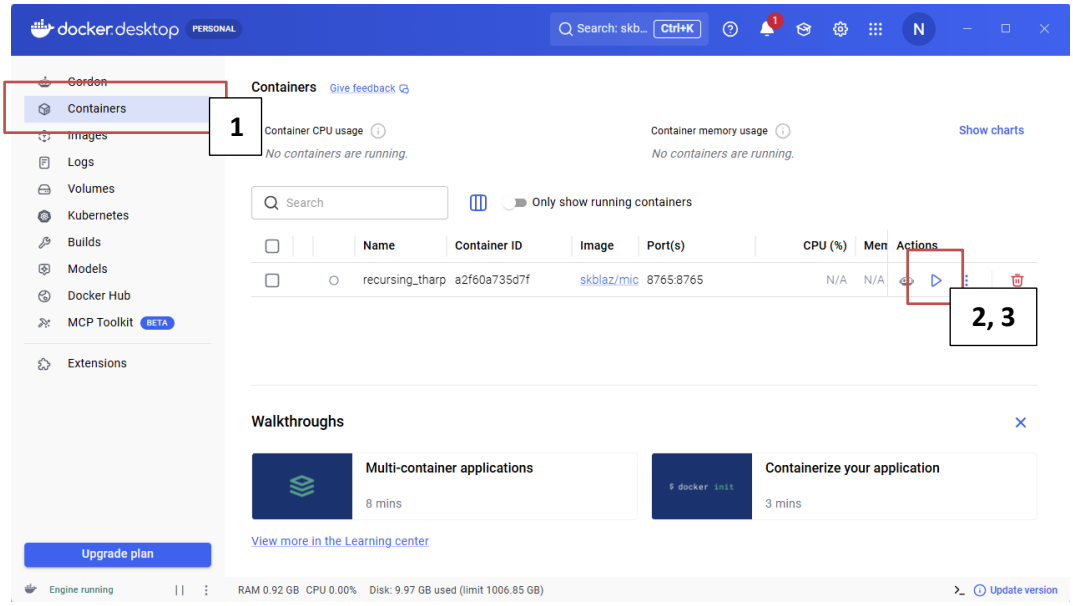
- Start the tool:

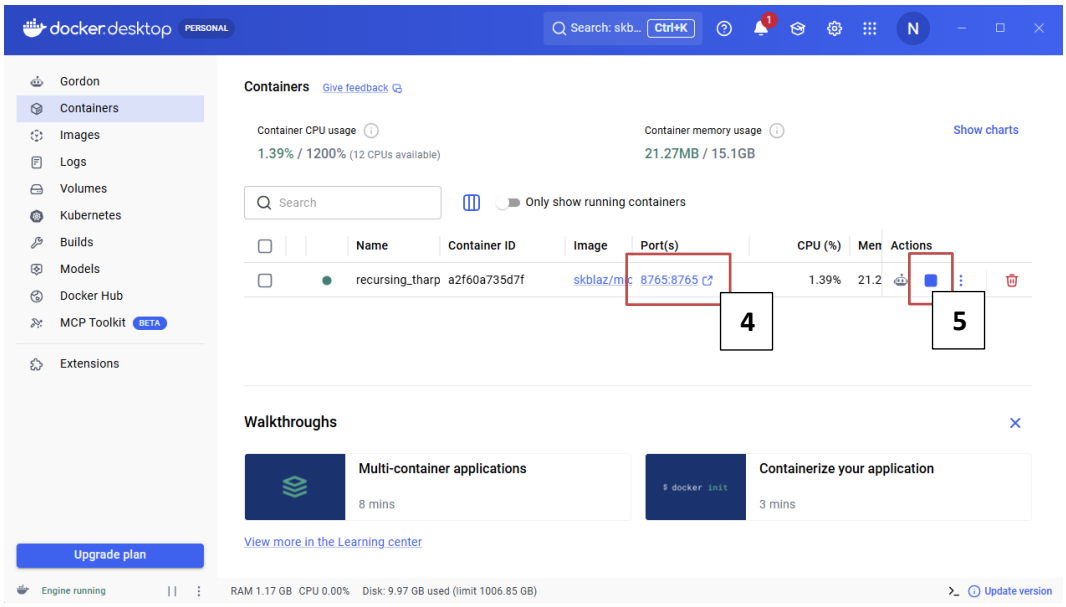
```
docker run -p 8765:8765 -v microics-data:/data skblaz/microics:latest
```

- Open http://localhost:8765 in your browser.
- To stop the container press on (■) below the Actions.

- **Restarting the existing container of MicroICS**
    1. Start Docker Desktop and go to Containers.
    2. Find your MicroICS container (it'll be stopped).
    3. Click the Start button (▶) next to it.
    4. Once it's running, click the 8765:8765 link, or open http://localhost:8765 in your browser.
    5. When you have finished the analyses, stop the container by pressing (■) below Actions.

**To get the latest version of the Docker image of MicroICS, you need to repeat the steps for setting up MicroICS (via Docker).**



## Installation on Linux

- **Install Docker**
  - Download and run Docker's official installation script:

```
curl -fsSL https://get.docker.com -o get-docker.sh
```

```
sudo sh get-docker.sh
```

  - Allow Docker to run without sudo:

By default, Docker requires sudo for every command, which causes permission problems with output files. Add your user to the Docker group so Docker can be run directly:

```
sudo usermod -aG docker $USER
```

  - Then reboot for this change to take effect:

```
sudo reboot
```

  - Check that Docker installed correctly (recommended)

```
docker --version
```

It should print a version number.

**Then continue to the "Installing MicroICS with Docker" steps — from that point on, Linux is identical to the Windows (pull the image, run it, open in the browser).**

## Installation on macOS

- **Install Docker**
  - Download Docker Desktop from https://docs.docker.com/desktop/setup/install/mac-install/ and install it. On Mac you may be offered an Apple chip version (M1–M4) or an Intel chip version – check which you have via → About This Mac → look at “Chip” or “Processor”. Install using the default options.
  - Start Docker Desktop: Open Docker Desktop from Applications and wait until the whale icon in the menu bar shows it is running. Docker must be running whenever you use MicroICS.
  - Check Docker is working (optional): Open Terminal and run:

```
docker --version
```

It should print a version number.

**Then continue to the "Installing MicroICS with Docker" steps — from that point on, Mac is identical to the Windows or Linux (pull the image, run it, open in the browser).**